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The cost of thinking is similar between large reasoning models and humans

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Do neural network models capture the cognitive demands of human reasoning? Across seven reasoning tasks, we show that the length of the chain-of-thought generated by large reasoning models predicts human reaction times both within tasks—tracking item-level difficulty—and across tasks—capturing broader differences in cognitive demands. This model-to-human alignment shows that out-of-the-box reasoning models reflect core features underlying problem and task complexity in human cognition, without requiring any built-in symbolic mechanisms.

reasoning | large reasoning models | chain-of-thought | cognitive modeling

Neural network models have ushered in a new era in cognitive science and neuroscience. Large language models (LLMs) and convolutional neural networks (CNNs) have proven tremendously useful as in silico model organisms for language and vision, respectively. Not only do they excel at the relevant behaviors, but they also capture the cognitive costs of perceptual processing and incremental language comprehension (1, 2). Yet, whether neural network models can achieve robust reasoning performance and thus serve as models of human reasoning remains debated, with many arguing that higher-level cognition would require additional or distinct mechanisms—potentially involving symbolic representations—that current neural networks may not possess (3, 4).

Previous attempts to use LLMs as models of reasoning have been mixed. Although LLMs display several human-like behaviors, including susceptibility to content effects (5) and verbal-thinking impairments (6), they often fail to engage in systematic exploration or causal reasoning, and small surface perturbations can yield large differences in output (7). One approach to improve the cognitive plausibility of LLMs as models of reasoning has been to train models directly on human behavioral data (8). This approach improves the alignment with human behavior but introduces a circularity: modeling behavior by training on the very same kind of behavior one seeks to explain, i.e., human responses in psychological experiments. Developing non-self-referential accounts of behavior that are grounded in independent principles, therefore, remains an important goal.

Recently, a new class of so-called 'large reasoning models' (LRMs) has emerged as a promising alternative to traditional LLMs. These models are derived from "standard" LLMs by additionally training them to solve tasks through step-by-step reasoning, often guided by reinforcement learning with verifiable rewards (RLVR, (9); note that this procedure differs from reinforcement learning from human feedback, or RLHF, which rewards outputs that align with human preferences for style and tone instead of correct reasoning). Although standard LLMs can also generate reasoning steps when prompted, LRMs generate extended chains of thought (CoT) by default, re-elaborating task information through intermediate steps (Fig. 1*A*), and achieve superior reasoning performance (9). Importantly, even though LRMs use language to generate chains of thought, the underlying computations are likely nonlinguistic—as in humans (10)—given that CoT text often misrepresents the true internal processes (11). Nevertheless, we hypothesized that CoT length could be a useful proxy of computational effort.

Building on this hypothesis, we tested whether the step-by-step computation in LRMs mirrors the processing effort humans expend when reasoning. Specifically, we asked whether the number of reasoning tokens generated by LRMs predicts the time humans take to solve problems across diverse reasoning domains.

Results

We evaluated an open-weights LRM [DeepSeek-R1, henceforth R1 (9)] on seven reasoning tasks: numeric arithmetic, verbal arithmetic, syllogisms, formal logic problems, relational reasoning, intuitive reasoning, and abstract structure mapping (H-ARC; see Fig. 1*B* for sample problems).

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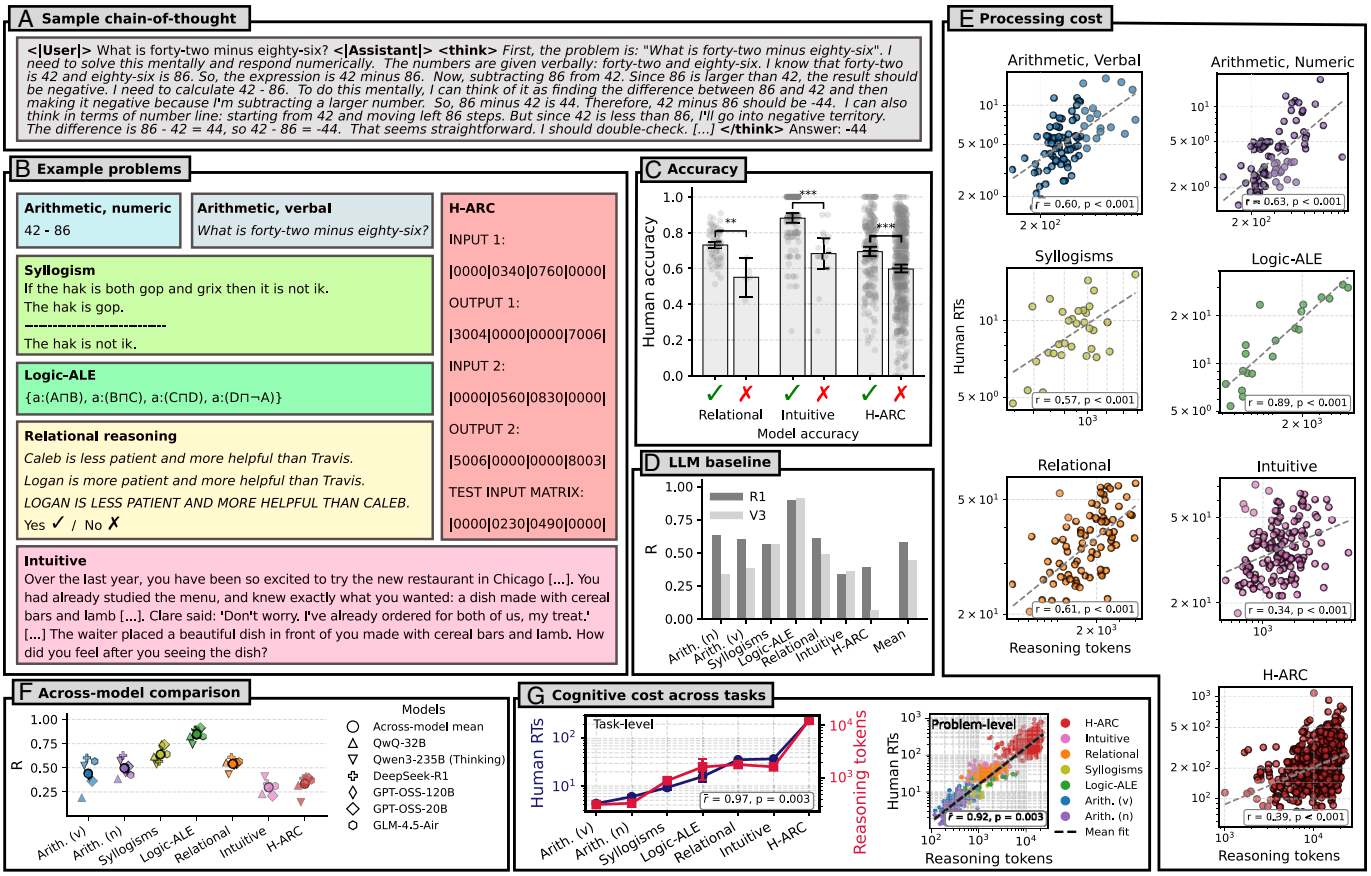

Fig. 1. (*A*) Sample chain-of-thought produced by R1. (*B*) Example reasoning problems. (*C*) Human accuracy scores (*y*-axis) grouped by items where R1 successfully solved the task (✓) or not (×). Because R1 often performed at ceiling, Fig. 1*C* includes only three datasets, where both human and model accuracy varied across items. (*D*) Comparison between the correlations with human response times obtained by R1 and its "base" homolog LLM DeepSeek-V3. (*E*) Relationship between the number of reasoning tokens used by R1 to solve a problem (*x*-axis) and human reaction times on the same problem (*y*-axis) within the various reasoning domains. Both axes are log-scaled. (*F*) Results obtained with six LRMs. (*G*) Relationship between reasoning token usage and reaction times across tasks. The line plot on the left reports the by-task averages of both token usage (red) and RTs (blue), whereas the scatter plot on the right depicts individual trials, colored by task.

R1 achieved higher accuracy on items that humans tended to solve correctly (Fig. 1*C*). To assess whether R1 also captures the *cognitive costs* associated with reasoning, we analyzed its reasoning traces—specifically, the number of reasoning tokens generated for each problem, and compared them to human reaction times (RTs). We found that the number of reasoning tokens generated by R1 reliably predicted human RTs within each task, with correlations ranging from $r = 0.34$ to $r = 0.89$ (mean across tasks: $\bar{r} = 0.57$; all $ps < 0.001$; Fig. 1*E*). (Differences among tasks in correlation strength are plausibly due to differences in data reliability; *SI Appendix*, section 2.) Problem length itself was also correlated with human RTs, but controlling for it via partial correlation (residualizing both CoT lengths and RTs) still yielded significant effects (partial mean $\bar{r} = 0.43$, all $ps < 0.05$; *SI Appendix*, section 4). This finding suggests that the model's computation reflects the relative cognitive cost experienced by human participants. Importantly, 4 of the 7 datasets (numeric and verbal arithmetic, syllogisms, intuitive reasoning) are either unpublished or were released after R1's training, excluding the possibility that the alignment with human behavior might be due to memorization, although, of course, similar problems may well have been present in the training data.

The alignment generalized to six LRMs, with small intermodel differences (Fig. 1*F* and *SI Appendix*, section 4). Notably, the effect was substantially weaker for R1's base model, DeepSeek-V3 ($\bar{r} = 0.44$ vs. $\bar{r} = 0.57$ for R1; $z = 4.39$, $p < 0.001$; Fig. 1*D*), indicating that reasoning-optimized training increases the model's alignment with human processing effort, although this advantage was not ubiquitous across tasks.

A general model of human reasoning should not only account for problem difficulty within tasks but also capture differences in difficulty *across* tasks that vary in problem structure and complexity. We found that the average number of reasoning tokens generated by R1 also captures average human RTs across the seven tasks, suggesting that the model mirrors broad differences in cognitive demand ($r = 0.97$, $P = 0.003$; Fig. 1 *G*, *Left*; this effect was not driven by problem length, *SI Appendix*, section 4). Moreover, we pooled trial-level responses from all tasks to test whether a single token-based measure could explain human RTs across diverse problem types. When controlling for dataset size via repeated subsampling, reasoning token counts consistently predicted item-level RTs ($\bar{r} = 0.92$, $P = 0.003$; Fig. 1 *G*, *Right*), demonstrating that R1's reasoning costs scale with human effort both within and across domains.

Discussion

Our findings suggest that LRMs exhibit strong alignment with human reasoning behavior, not only in terms of which problems are solved correctly but also in the processing cost associated with solving them. This finding challenges the long-held view that explicitly symbolic architectures are necessary to support human-like thinking (3, 4). The models we evaluate do not have any abstract inference rules built-in, and yet they closely match human reasoning

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effort. Instead, our finding aligns with approaches that model higher-level cognition as the product of goal-directed optimization (here, correctly solving reasoning problems) under environmental and computational constraints (12). At an abstract level, LRMs' learning trajectories also resemble those of humans: During pre-training, the models observe a wide range of correct reasoning examples, similar to how students learn from textbooks or instruction; during reinforcement learning, they receive feedback on their performance, akin to practicing and learning from success or failure. After training, the models allocate more effort to harder problems—a hallmark of efficient problem-solving. Future work should build on these findings to both evaluate model-to-human similarity in other domains of reasoning and to develop and evaluate hypotheses about the reasons for the observed alignment.

Methods

Datasets. Our suite of datasets spanned seven reasoning tasks. The first three were based on unpublished data from the Fedorenko lab and received approval from MIT's Committee On the Use of Humans as Experimental Subjects (protocol number 2010000243) with informed written consent from all participants. In the first two tasks, participants solved arithmetic problems (addition/subtraction) presented in numeric or verbal format. In the third task, participants solved three-sentence syllogisms (modus ponens/tollens) where the problems used either real words or nonwords. In the fourth task (13), participants with prior logic experience judged the consistency of 20 logic problems in $\mathcal{A}$LE format. In the fifth task (14), participants solved spatial and nonspatial relational reasoning problems. In the sixth task (15), participants solved intuitive reasoning problems spanning both social and physical content. In the seventh task (16), participants solved transformation-based grid problems. Reliability was high across datasets (split-half ρ between 0.60 and 0.93; *SI Appendix*, section 2). (See *SI Appendix*, section 1 for dataset details.)

Model Evaluation. In our main analyses, we tested DeepSeek-R1, an LRM, and its "base" LLM counterpart DeepSeek-V3 (see Fig. 1*F* and *SI Appendix*, section 4 for a replication with six LRMs). R1 outputs a reasoning trace followed by a final answer, where the two are separated by a special token (Fig. 1*A*). Reasoning effort was measured by counting subword tokens within the reasoning trace. For V3, we appended the chain-of-thought prompt "Let's think step by step" to the problem text and used the full output length.

Analysis. We calculated Pearson correlations between log-transformed number of reasoning tokens and average human reaction times, both within and across tasks. Model-human alignment across tasks (Fig. 1 *G*, *Right*) was assessed using bootstrap-resampled trial-level data (*SI Appendix*, section 4).

Data, Materials, and Software Availability. All code and data are available on OSF (https://osf.io/r3kum/) (17). This article uses data from multiple pre-existing studies; the relevant citations are reported in §Datasets and described in detail in *SI Appendix*, section 1.

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